MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **CBSE CLASS 10 – ASSESSMENT TEST** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 11: Areas Related to Circles** | **MM:** 20

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

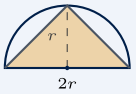
Official Evaluation Key | Maths Manthan Meerut

SECTION A: MULTIPLE CHOICE QUESTIONS MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. Area of the largest triangle that can be inscribed in a semi-circle of radius r units is: [1]

(A) r^2 sq. units (B) $\frac{1}{2}r^2$ sq. units (C) $2r^2$ sq. units (D) $\sqrt{2}r^2$ sq. units

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]



Step 1: Base and Maximum Altitude [0.5 Mark]

The base of any triangle inscribed in a semicircle cannot exceed the diameter:

$$\text{Base } b = 2r$$

The maximum altitude from the circle boundary to the diameter base occurs at the apex perpendicular to the centre:

$$\text{Maximum Height } h = r$$

Step 2: Area Calculation [0.5 Mark]

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 2r \times r = r^2 \text{ sq. units}$$

Correct Option: (A) r^2 sq. units

Maximum Triangle Area = $\mathbf{r^2}$ sq. units

Q2. If the perimeter of a circle is equal to that of a square, then the ratio of their areas is: [1]

(A) 22 : 7 (B) 14 : 11 (C) 7 : 22 (D) 11 : 14

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Equating Perimeters [0.5 Mark]

Let the radius of the circle be r and the side of the square be a.

$$\text{Circumference of circle} = \text{Perimeter of square} \implies 2\pi r = 4a \implies a = \frac{\pi r}{2}$$

Step 2: Ratio of Area of Circle to Area of Square [0.5 Mark]

$$\frac{\text{Area(circle)}}{\text{Area(square)}} = \frac{\pi r^2}{a^2} = \frac{\pi r^2}{(\pi r/2)^2} = \frac{\pi r^2}{\pi^2 r^2/4} = \frac{4}{\pi} = \frac{4}{22/7} = \frac{28}{22} = \frac{14}{11}$$

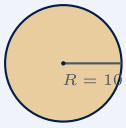
Correct Option: (B) 14 : 11

$$\boxed{\frac{\text{Area of circle}}{\text{Area of square}} = \mathbf{14 : 11}}$$

Q3. It is proposed to build a single circular park equal in area to the sum of areas of two circular parks of diameters 16 m and 12 m in a locality. The radius of the new park would be: [CBSE 2012] [1]

(A) 10 m (B) 15 m (C) 20 m (D) 24 m

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]



Step 1: Radii of Existing Parks [0.5 Mark]

$$r_1 = \frac{16}{2} = 8 \text{ m}, \qquad r_2 = \frac{12}{2} = 6 \text{ m}$$

Step 2: Conservation of Area [0.5 Mark]

Let the radius of the combined new park be R.

$$\pi R^2 = \pi r_1^2 + \pi r_2^2 \implies R^2 = 8^2 + 6^2 = 64 + 36 = 100$$

$$R = \sqrt{100} = 10 \text{ m}$$

Correct Option: (A) 10 m

$$\boxed{\text{Radius of the new park } R = \mathbf{10} \text{ m}}$$

Q4. If the areas of two circles are in the ratio 4 : 9, then the ratio of the perimeters of their semi-circles is: [1]

(A) 2 : 3 (B) 3 : 2 (C) 1 : 2 (D) 1 : 3

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Ratio of Radii [0.5 Mark]

$$\frac{\text{Area}_1}{\text{Area}_2} = \frac{\pi r_1^2}{\pi r_2^2} = \frac{4}{9} \implies \frac{r_1}{r_2} = \sqrt{\frac{4}{9}} = \frac{2}{3}$$

Step 2: Ratio of Semicircular Perimeters [0.5 Mark]

The perimeter of a semicircle with radius r is $P = \pi r + 2r = r(\pi + 2)$.

$$\frac{P_1}{P_2} = \frac{r_1(\pi + 2)}{r_2(\pi + 2)} = \frac{r_1}{r_2} = \frac{2}{3} = 2 : 3$$

Correct Option: (A) 2 : 3

$$\boxed{\text{Ratio of perimeters of their semicircles} = \mathbf{2 : 3}}$$

Q5. If the difference between the circumference and the radius of a circle is 37 cm. If $\pi = \frac{22}{7}$, then the circumference (in cm) of the circle is: [1]

(A) 15.4 (B) 44 (C) 14 (D) 7

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Formulating and Solving for Radius r [0.5 Mark]

$$C - r = 37 \implies 2\pi r - r = 37 \implies r(2\pi - 1) = 37$$

$$r\left(2 \times \frac{22}{7} - 1\right) = 37 \implies r\left(\frac{44-7}{7}\right) = 37 \implies r\left(\frac{37}{7}\right) = 37 \implies r = 7 \text{ cm}$$

Step 2: Finding Circumference [0.5 Mark]

$$C = 2\pi r = 2 \times \frac{22}{7} \times 7 = 44 \text{ cm}$$

Correct Option: (B) 44

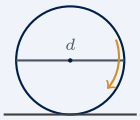
Circumference of the circle = **44** cm

SECTION B: Short Answer Type-I

MATHS MANTHAN | **(3 × 2 = 6 Marks)**

Q6. A bicycle wheel makes 5000 revolutions in moving 11 km. Find the diameter of the wheel. [2]

▶ **STEP-BY-STEP SOLUTION:** [2 Marks]



Step 1: Circumference of Wheel [1 Mark]

Total distance $D = 11 \text{ km} = 11 \times 1000 \times 100 = 1100000$ cm.

$$\text{Distance in 1 revolution } C = \frac{\text{Total Distance}}{n} = \frac{1100000}{5000} = 220 \text{ cm}$$

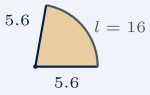
Step 2: Diameter Evaluation [1 Mark]

$$C = \pi d \implies \frac{22}{7}d = 220 \implies d = \frac{220 \times 7}{22} = 10 \times 7 = 70 \text{ cm} = 0.7 \text{ m}$$

Diameter of the wheel = **70** cm (or **0.7** m)

Q7. The perimeter of a sector of a circle of radius 5.6 cm is 27.2 cm. Find the area of the sector. [2]

▶ **STEP-BY-STEP SOLUTION:** [2 Marks]



Step 1: Finding Arc Length l [1 Mark]

Radius $r = 5.6$ cm. Perimeter of sector $= l + 2r = 27.2$ cm.

$$l + 2(5.6) = 27.2 \implies l + 11.2 = 27.2 \implies l = 27.2 - 11.2 = 16 \text{ cm}$$

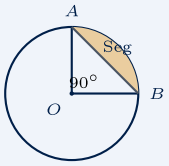
Step 2: Area of the Sector [1 Mark]

$$\text{Area} = \frac{1}{2}lr = \frac{1}{2} \times 16 \times 5.6 = 8 \times 5.6 = 44.8 \text{ cm}^2$$

Area of the sector = **44.8** cm^2

Q8. A chord AB of a circle of radius 10 cm subtends a right angle at the centre. Find the area of the major and minor segments. [Take $\pi = 3.14$] [NCERT] **[2]**

▶ **STEP-BY-STEP SOLUTION:** [2 Marks]



Step 1: Area of Minor Segment [1 Mark]

Radius $r = 10$ cm, central angle $\theta = 90°$.

$$\text{Area}_{\text{minor sector}} = \frac{90°}{360°} \times 3.14 \times 10^2 = \frac{1}{4} \times 314 = 78.5 \text{ cm}^2$$

$$\text{Area}(\triangle OAB) = \frac{1}{2} \times 10 \times 10 = 50 \text{ cm}^2$$

$$\text{Area}_{\text{minor segment}} = 78.5 - 50 = 28.5 \text{ cm}^2$$

Step 2: Area of Major Segment [1 Mark]

$$\text{Total Area of circle} = \pi r^2 = 3.14 \times 100 = 314 \text{ cm}^2$$

$$\text{Area}_{\text{major segment}} = 314 - 28.5 = 285.5 \text{ cm}^2$$

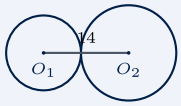
Minor Segment Area = **28.5** cm^2, Major Segment Area = **285.5** cm^2

SECTION C: Short Answer Type-II

MATHS MANTHAN | **(3 × 3 = 9 Marks)**

Q9. Two circles touch externally. The sum of their areas is 130π sq cm and the distance between their centres is 14 cm. Find the radii of the circles. **[3]**

▶ **STEP-BY-STEP SOLUTION:** [3 Marks]



Step 1: Setting up System of Equations [1 Mark]

Let the radii of the two circles be r_1 and r_2. Since the circles touch externally:

$$r_1 + r_2 = 14 \implies r_2 = 14 - r_1$$

$$\text{Sum of areas} = \pi r_1^2 + \pi r_2^2 = 130\pi \implies r_1^2 + r_2^2 = 130$$

Step 2: Quadratic Formation and Solution [1.5 Marks]

$$r_1^2 + (14 - r_1)^2 = 130 \implies r_1^2 + 196 - 28r_1 + r_1^2 = 130$$

$$2r_1^2 - 28r_1 + 66 = 0 \implies r_1^2 - 14r_1 + 33 = 0$$

$$(r_1 - 11)(r_1 - 3) = 0 \implies r_1 = 11 \text{ cm} \quad \text{or} \quad r_1 = 3 \text{ cm}$$

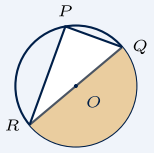
Step 3: Conclusion [0.5 Mark]

The radii of the two circles are 11 cm and 3 cm.

Radii of the two circles are **11** cm and **3** cm

Q10. Find the area of the shaded region in the given figure, if $PR = 24$ cm, $PQ = 7$ cm and O is the centre of the circle. [CBSE 2008] **[3]**

▶ **STEP-BY-STEP SOLUTION:** [3 Marks]



Step 1: Hypotenuse / Diameter Calculation [1 Mark]

Since RQ passes through centre O, RQ is a diameter. Hence, $\angle RPQ = 90°$ (angle in a semicircle).

$$\text{Diameter } RQ = \sqrt{PR^2 + PQ^2} = \sqrt{24^2 + 7^2} = \sqrt{576 + 49} = \sqrt{625} = 25 \text{ cm}$$

$$\text{Radius } r = \frac{25}{2} \text{ cm}$$

Step 2: Area of Semicircle and Triangle [1 Mark]

$$\text{Area}_{\text{semicircle}} = \frac{1}{2}\pi r^2 = \frac{1}{2} \times \frac{22}{7} \times \left(\frac{25}{2}\right)^2 = \frac{11 \times 625}{28} = \frac{6875}{28} \text{ cm}^2$$

$$\text{Area}(\triangle RPQ) = \frac{1}{2} \times PR \times PQ = \frac{1}{2} \times 24 \times 7 = 84 \text{ cm}^2$$

Step 3: Area of Shaded Region [1 Mark]

$$\text{Area} = \frac{6875}{28} - 84 = \frac{6875 - 2352}{28} = \frac{4523}{28} \approx 161.54 \text{ cm}^2$$

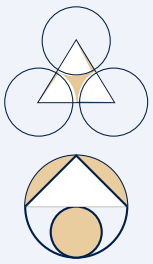
$$\boxed{\text{Area of the shaded region} = \frac{4523}{28} \text{ cm}^2 \approx \mathbf{161.54} \text{ cm}^2}$$

Q11. The area of an equilateral triangle is 17320 cm^2. With each vertex of the triangle a circle is drawn with radius equal to half the length of the side of the triangle in the given figure. Find the area of the shaded region. [Take $\sqrt{3} = 1.732$] **[3]**

OR

In the given figure, AB and CD are two diameters of a circle (with centre O) perpendicular to each other and OD is the diameter of the smaller circle. If $OA = 7$ cm, find the area of the shaded region. [NCERT] **[3]**

▶ **STEP-BY-STEP SOLUTION:** [3 Marks]



MAIN QUESTION: Equilateral Triangle Shaded Region

• Let side length of equilateral triangle be a:

$$\text{Area} = \frac{\sqrt{3}}{4}a^2 = 17320 \implies \frac{1.732}{4}a^2 = 17320 \implies a^2 = 40000 \implies a = 200 \text{ cm}$$

• Radius of each circular sector $r = \frac{a}{2} = 100$ cm, $\theta = 60°$:

$$\text{Area of 3 sectors} = 3\times\left(\frac{60°}{360°} \times \pi r^2\right) = \frac{1}{2}\pi r^2 = \frac{1}{2}\times 3.14 \times 100^2 = 15700 \text{ cm}^2$$

$$\text{Area of shaded region} = 17320 - 15700 = 1620 \text{ cm}^2$$

OR QUESTION: Perpendicular Diameters Shaded Region

• Larger circle radius $R = OA = 7$ cm. Small circle diameter $= OD = 7$ cm $\implies r = 3.5$ cm.

$$\text{Area of small circle} = \pi r^2 = \frac{22}{7} \times \left(\frac{7}{2}\right)^2 = \frac{77}{2} = 38.5 \text{ cm}^2$$

• In semicircle ACB: $\text{Area}_{\text{semicircle}} = \frac{1}{2}\pi R^2 = \frac{1}{2} \times \frac{22}{7} \times 49 = 77 \text{ cm}^2$.

$$\text{Area}(\triangle ABC) = \frac{1}{2} \times AB \times OC = \frac{1}{2} \times 14 \times 7 = 49 \text{ cm}^2$$

$$\text{Area of 2 segments} = 77 - 49 = 28 \text{ cm}^2$$

$$\text{Total shaded area} = 38.5 + 28 = 66.5 \text{ cm}^2$$

Main Answer: 1620 cm²	**OR Answer: 66.5 cm²**

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